

TARNOW, Poland

WMO: 125750

Lat: 50.0299N

Lon: 20.9839E

Elev: 209

StdP: 98.84

Time Zone: 1.00 (EUC)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.5	-11.5	-18.2	0.8	-15.0	-14.6	1.1	-10.1	5.6	1.7	5.2	1.0	0.9	80	0.459

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	31.0	20.8	29.2	19.9	27.5	19.0	21.8	29.0	20.9	27.4	20.0	25.8	2.1	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.5	14.6	24.4	18.7	13.8	23.4	17.8	13.1	22.8	64.5	29.1	61.2	27.6	58.2	25.8	25.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
5.0	4.4	3.9	DB	-18.5	34.2	5.0	1.7	-22.1	35.4	-25.1	36.4	-27.9	37.4	-31.6	38.7	
			WB	-18.8	23.5	5.0	0.9	-22.4	24.1	-25.3	24.7	-28.1	25.2	-31.8	25.8	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.7	-1.3	-0.1	4.1	10.1	14.3	18.0	19.8	19.2	14.7	10.0	5.5	1.3
	DBStd	8.51	5.66	5.52	5.11	3.96	3.95	3.65	3.32	3.32	3.56	4.19	4.43	4.58
	HDD10.0	1354	352	283	190	45	8	0	0	0	4	55	146	271
	HDD18.3	3344	610	516	441	247	136	49	24	29	118	260	386	529
	CDD10.0	1233	0	1	7	49	141	241	302	285	144	53	9	1
	CDD18.3	184	0	0	0	1	10	41	68	56	8	0	0	0
	CDH23.3	1878	0	0	0	28	138	404	668	544	92	3	0	0
	CDH26.7	548	0	0	0	5	25	111	214	180	13	0	0	0
Wind	WSAvg	1.8	1.9	1.9	2.1	1.9	1.8	1.7	1.6	1.5	1.5	1.7	1.9	2.0
Precipitation	PrecAvg	719	36	31	37	51	81	105	107	89	56	42	43	41
	PrecMax	1024	92	89	82	119	164	207	294	173	103	191	84	75
	PrecMin	463	8	1	2	13	35	41	33	21	8	3	14	9
	PrecStd	142	20	17	19	28	32	43	59	39	26	35	16	18
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.2	14.1	19.6	26.8	29.2	32.5	33.7	33.2	28.2	23.6	17.3	12.3
		MCWB	8.0	8.2	11.0	15.3	18.2	21.7	21.7	21.1	18.7	15.8	12.4	8.5
	2%	DB	9.0	10.7	16.4	23.0	26.9	29.7	31.0	30.7	26.1	21.4	15.3	10.5
		MCWB	6.7	7.2	9.6	13.4	17.2	20.3	20.9	20.7	17.9	14.9	11.3	7.6
	5%	DB	7.2	8.9	14.0	20.6	24.6	27.4	29.0	28.6	23.7	19.3	13.6	9.0
		MCWB	5.0	5.8	8.4	12.3	16.7	19.0	19.9	20.0	16.6	14.1	10.4	6.5
	10%	DB	5.6	7.1	11.8	18.2	22.2	25.6	27.2	26.4	21.6	16.8	11.9	7.3
		MCWB	3.8	4.6	7.4	11.1	15.4	18.4	19.0	19.1	15.9	13.0	9.3	5.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.5	9.1	11.7	15.9	19.6	22.7	22.8	22.9	20.0	16.9	13.0	9.2
		MCDB	10.6	12.2	17.3	25.0	26.7	30.3	30.7	30.1	26.7	22.1	16.0	11.9
	2%	WB	6.9	7.3	10.2	14.2	18.1	21.3	21.8	21.5	18.5	15.4	11.9	7.8
		MCDB	8.7	10.4	15.5	21.2	24.9	27.9	29.0	28.7	24.1	20.1	14.8	10.1
	5%	WB	5.2	6.1	8.9	13.0	17.0	20.1	20.9	20.6	17.5	14.4	10.7	6.7
		MCDB	6.9	8.5	13.0	19.1	23.5	25.7	27.2	27.2	22.3	18.4	13.2	8.7
	10%	WB	4.0	4.9	7.6	12.0	15.9	19.0	20.0	19.7	16.5	13.3	9.4	5.5
		MCDB	5.5	6.9	11.2	17.3	21.4	24.1	25.5	25.3	20.5	16.6	11.5	7.2
Mean Daily Temperature Range	5% DB	MDBR	5.5	6.6	8.6	11.0	10.8	10.8	11.1	11.2	10.5	9.1	6.4	5.3
		MCDBR	6.5	9.4	12.6	15.7	14.8	14.1	14.4	14.6	14.3	13.3	9.0	6.8
		MCWBR	4.9	6.1	6.9	7.6	6.7	6.0	5.7	6.0	7.0	7.4	5.6	4.8
	5% WB	MCDBR	6.1	8.2	11.4	14.0	13.8	12.5	12.8	13.2	12.2	12.1	8.0	6.6
		MCWBR	4.8	5.7	6.6	7.0	6.7	5.6	5.5	5.7	6.3	7.0	5.2	4.9
Clear-Sky Solar Irradiance	taub	0.316	0.340	0.384	0.438	0.430	0.432	0.450	0.444	0.410	0.373	0.346	0.311	
	taud	2.278	2.242	2.212	2.116	2.193	2.224	2.186	2.203	2.259	2.338	2.348	2.335	
	Ebn at Noon	706	783	810	803	829	829	806	789	775	738	667	660	
	Edh at Noon	71	94	116	143	138	135	138	130	111	86	67	59	
All-Sky Solar Radiation	RadAvg	0.86	1.62	2.76	4.05	4.96	5.29	5.06	4.63	3.27	1.97	0.97	0.65	
	RadStd	0.14	0.26	0.35	0.47	0.60	0.45	0.60	0.36	0.47	0.28	0.12	0.09	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (34 neighbors)	+0.54	N/A	N/A	+0.90	+0.46	+0.40	-126	-170	+75	+30	

Nomenclature: See separate page